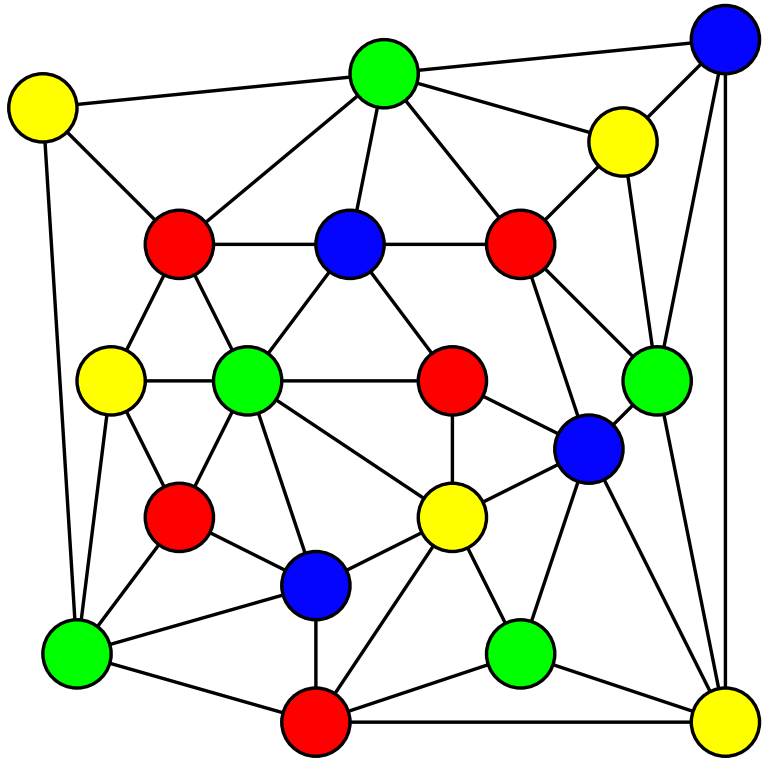
Algorithms and Analysis

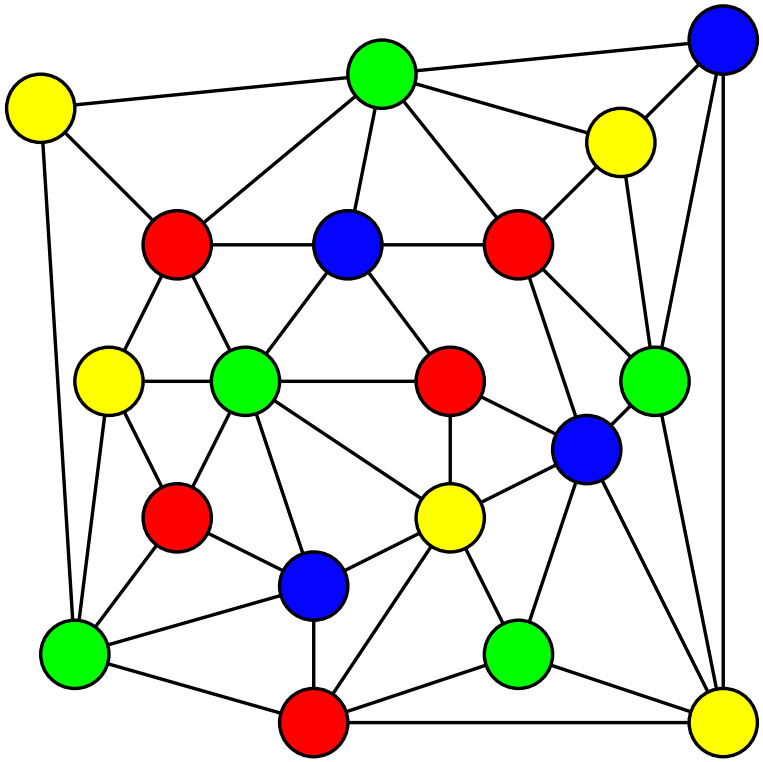
Lesson 17: *Think Graphically*



Graph theory, applications of graphs, graph problems

Outline

1. **Graph Theory**
2. Applications of Graphs
   - Geometric applications
   - Relational applications
3. Implementing Graphs
4. Graph Problems



Motivation

- Many different problems can be described in terms of graphs
- This often reveals the true nature of the problem
- It unifies many apparently different problems
- As much is known about graph problems it often provides a pointer to the solution

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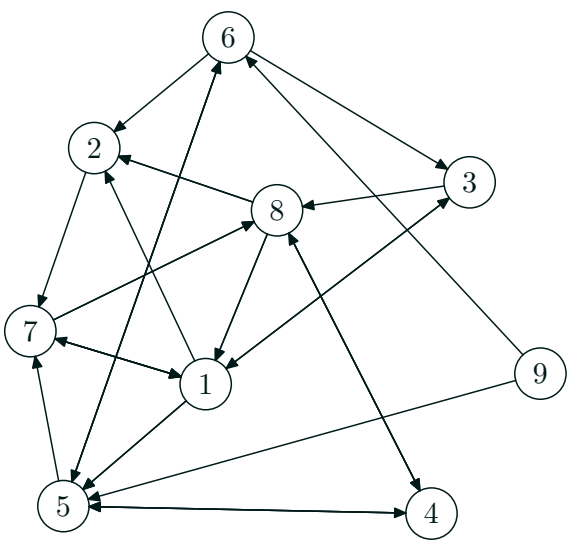
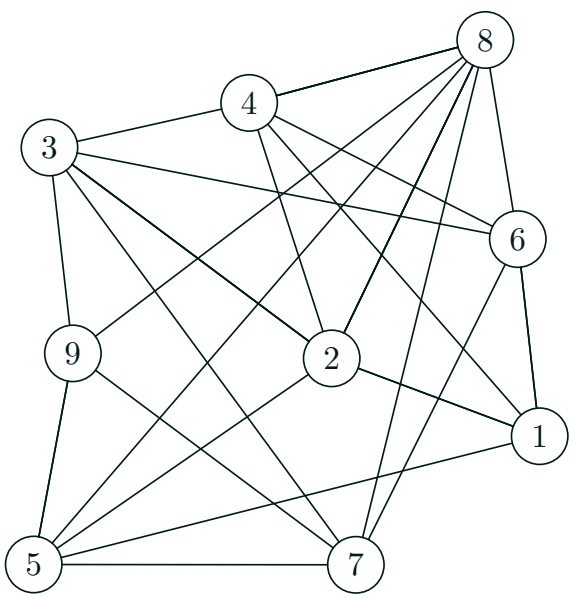
Definition of a Graph

- A graph, G, can be described by
  - ⋆ A set of vertices or nodes $\mathcal{V} = \{1, 2, 3 \ldots n\}$
  - ⋆ A set of edges $\mathcal{E} = \{(i, j) | \text{vertex } i \text{ is connected to vertex } j\}$
- The edges may be
  - ⋆ **directed**—sometimes called a **digraph**
  - ⋆ **undirected**





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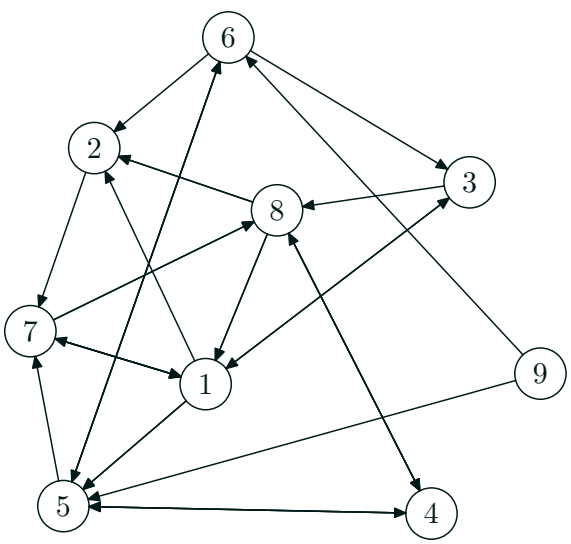

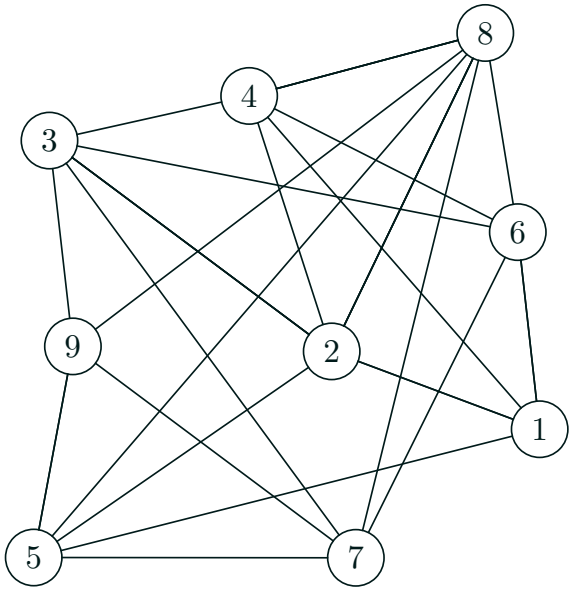

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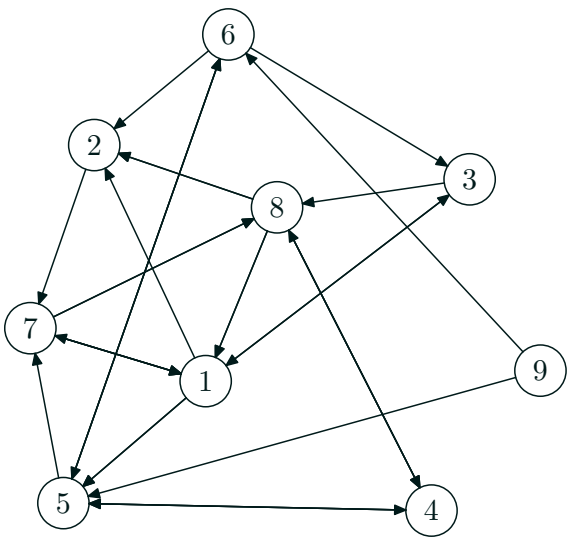

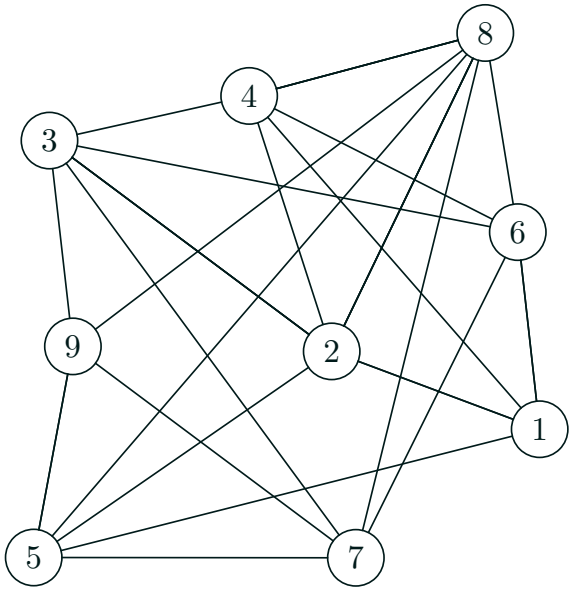

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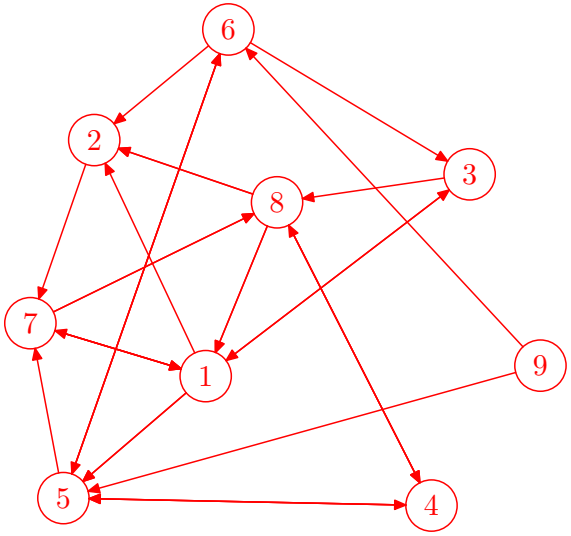

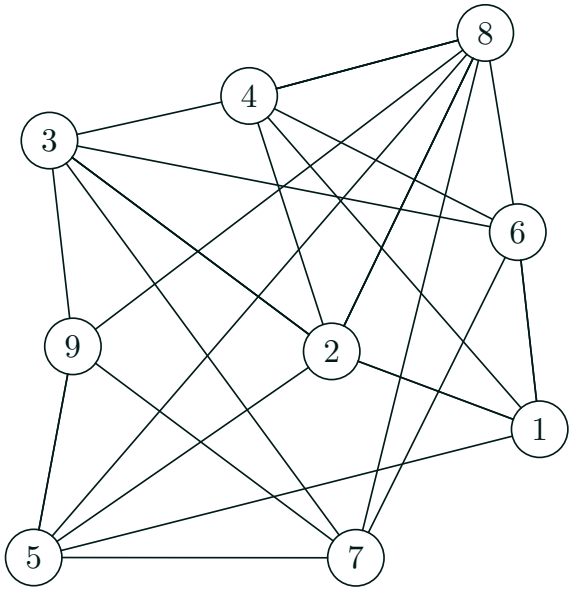

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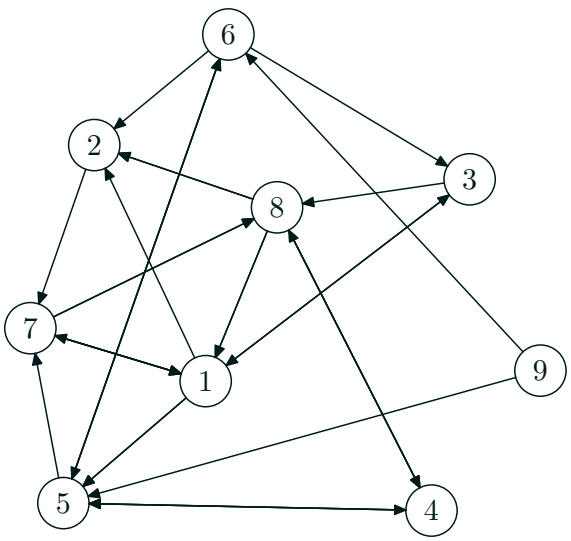

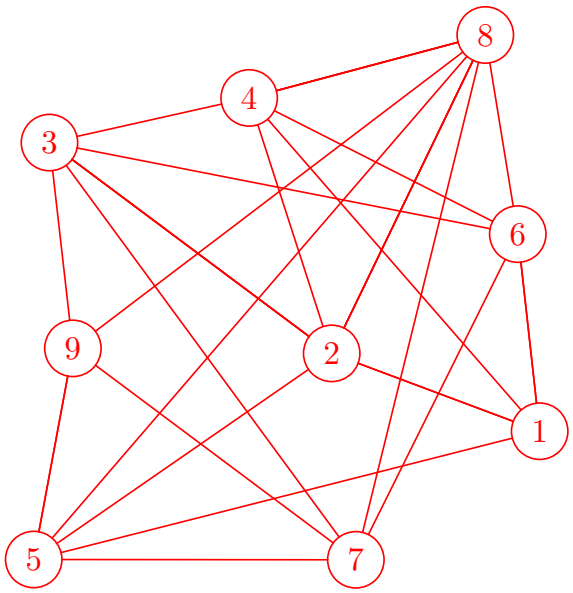

Connected and Unconnected Graphs

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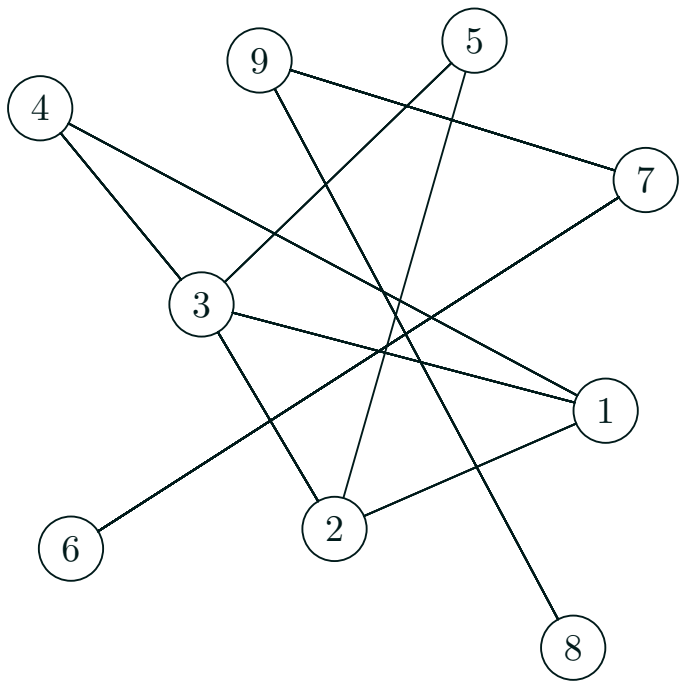

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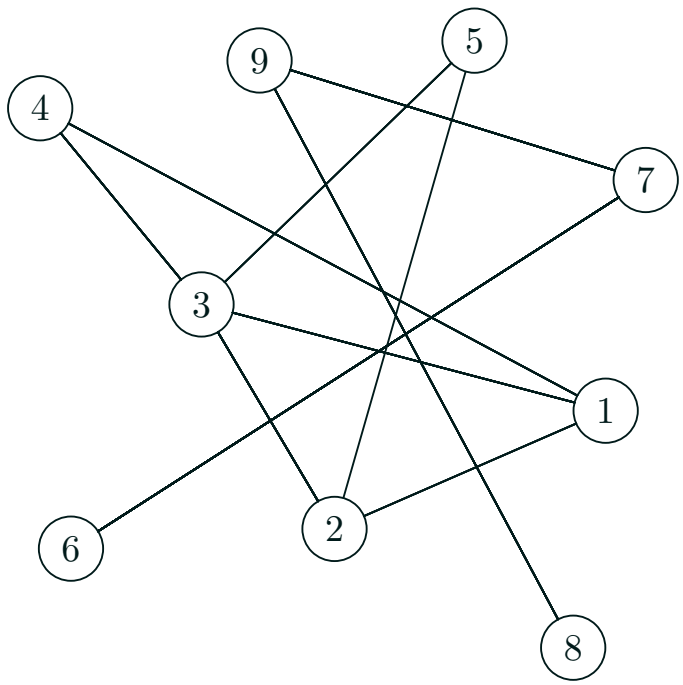

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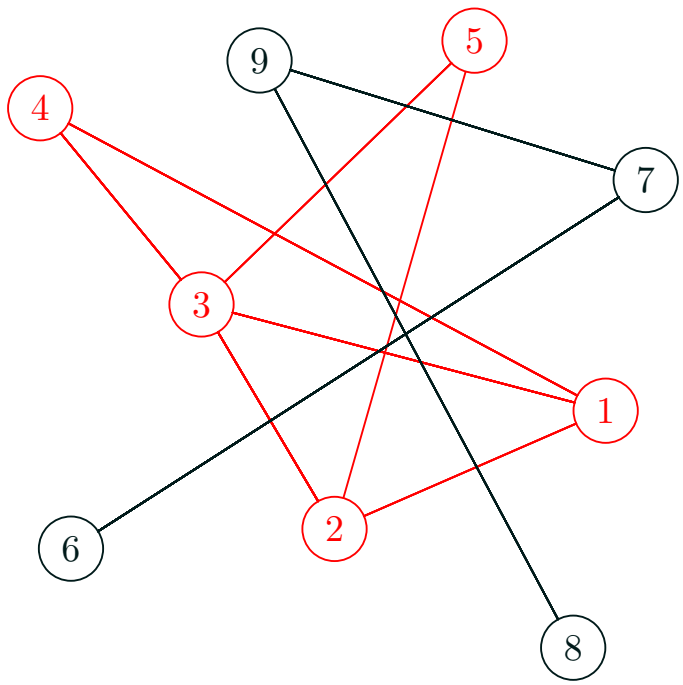

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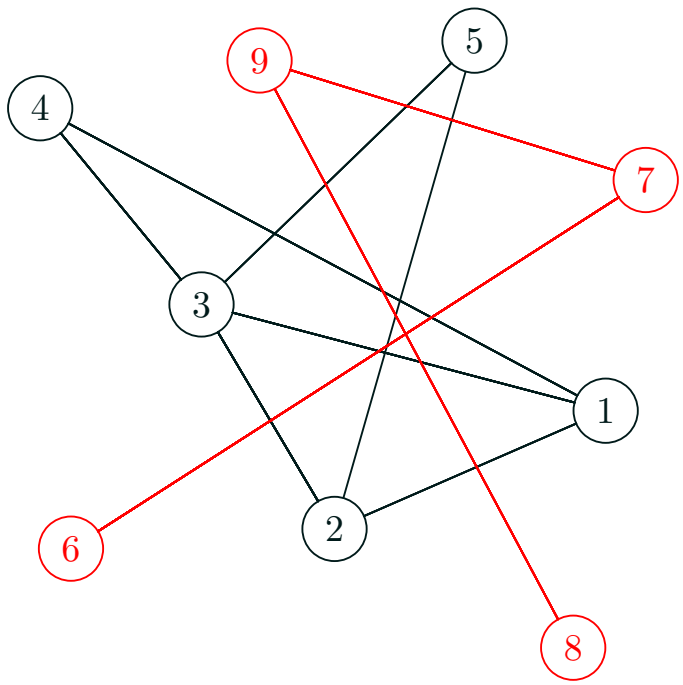
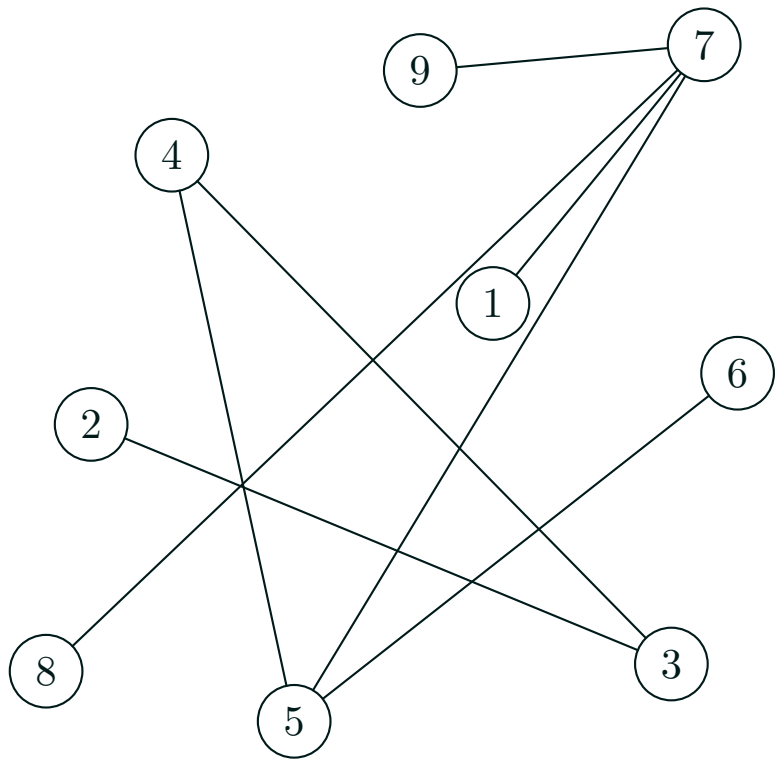


Trees

- A tree is a connected graphs with no cycles
- A tree will have $n - 1$ edges



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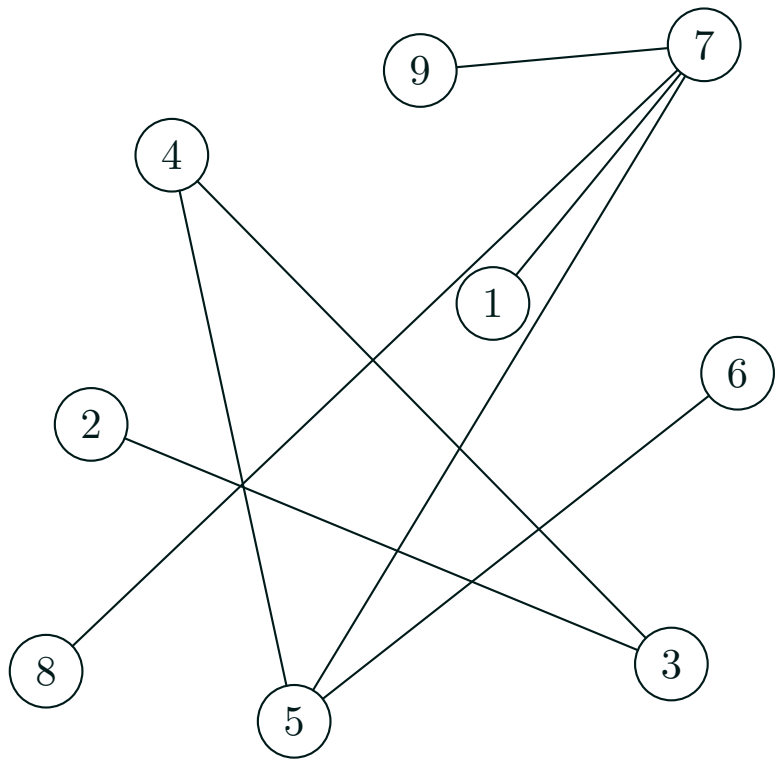
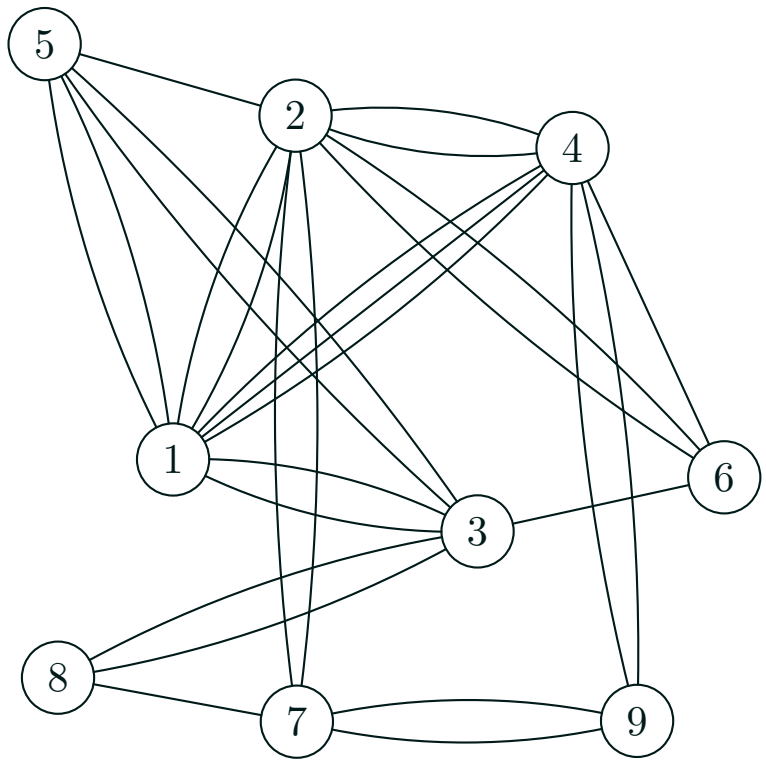


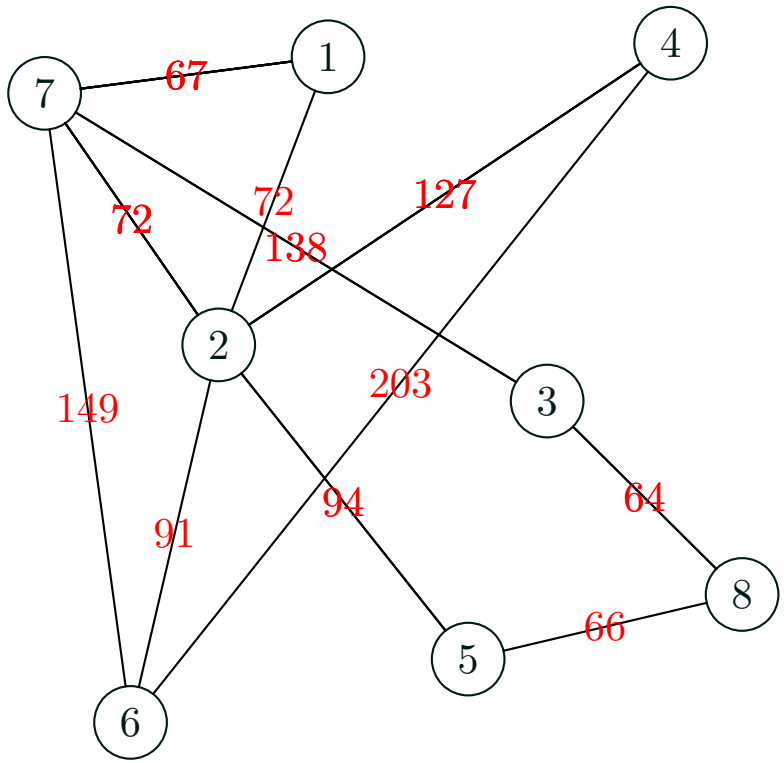
Multigraphs

- If the collection of edges is a *multiset* then we obtain a **multigraphs** where more than one edge is allowed between pairs of vertices



Weighted Graphs

- If we assign a number to an edge we obtain a **weighted graph**



Networks

- Sometimes we add more information to the graph
- E.g. attributes to the nodes or edges
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   - Geometric applications
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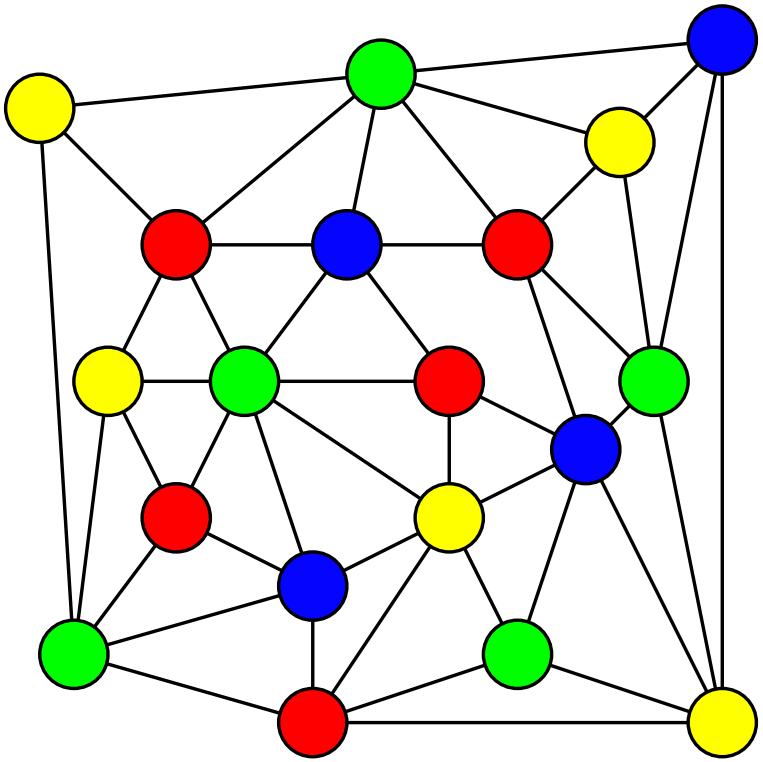
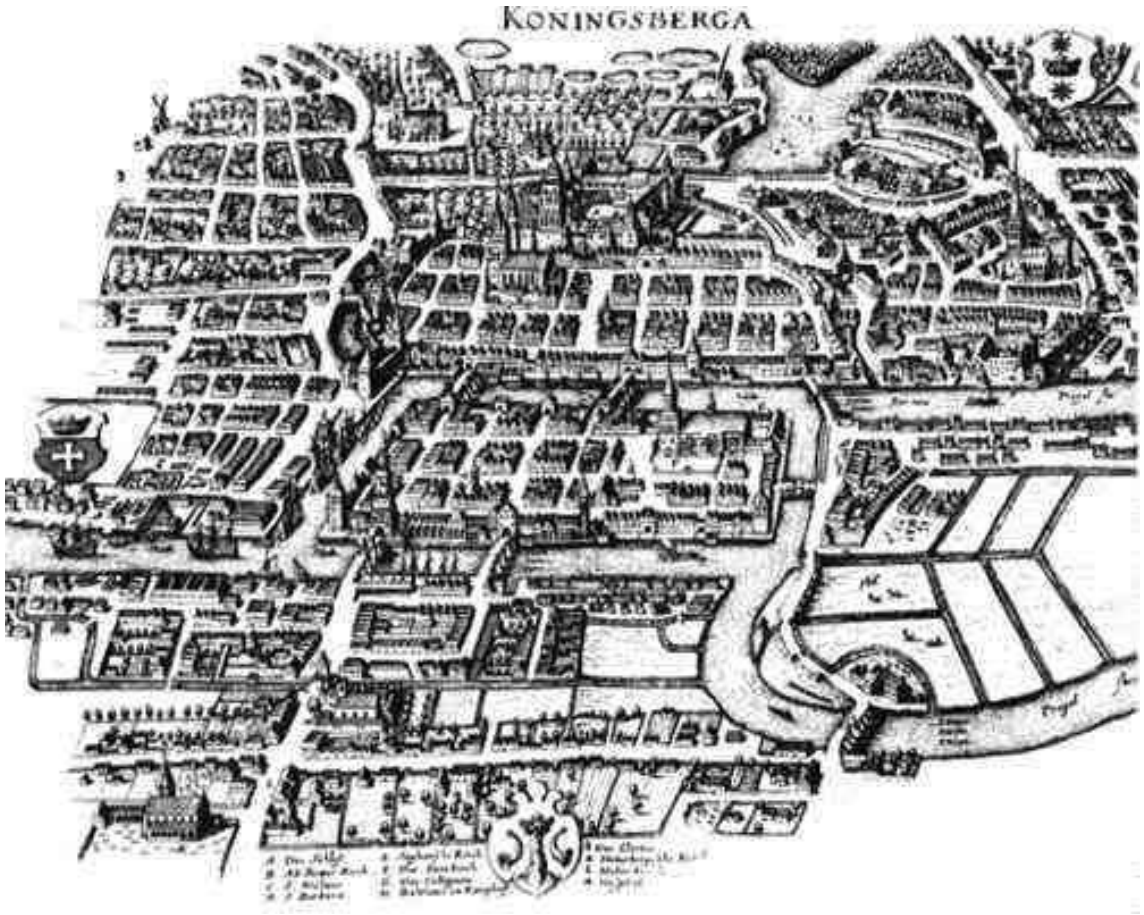


Bridges of Königsberg

Is there a tour around Königsberg going over every bridge once?



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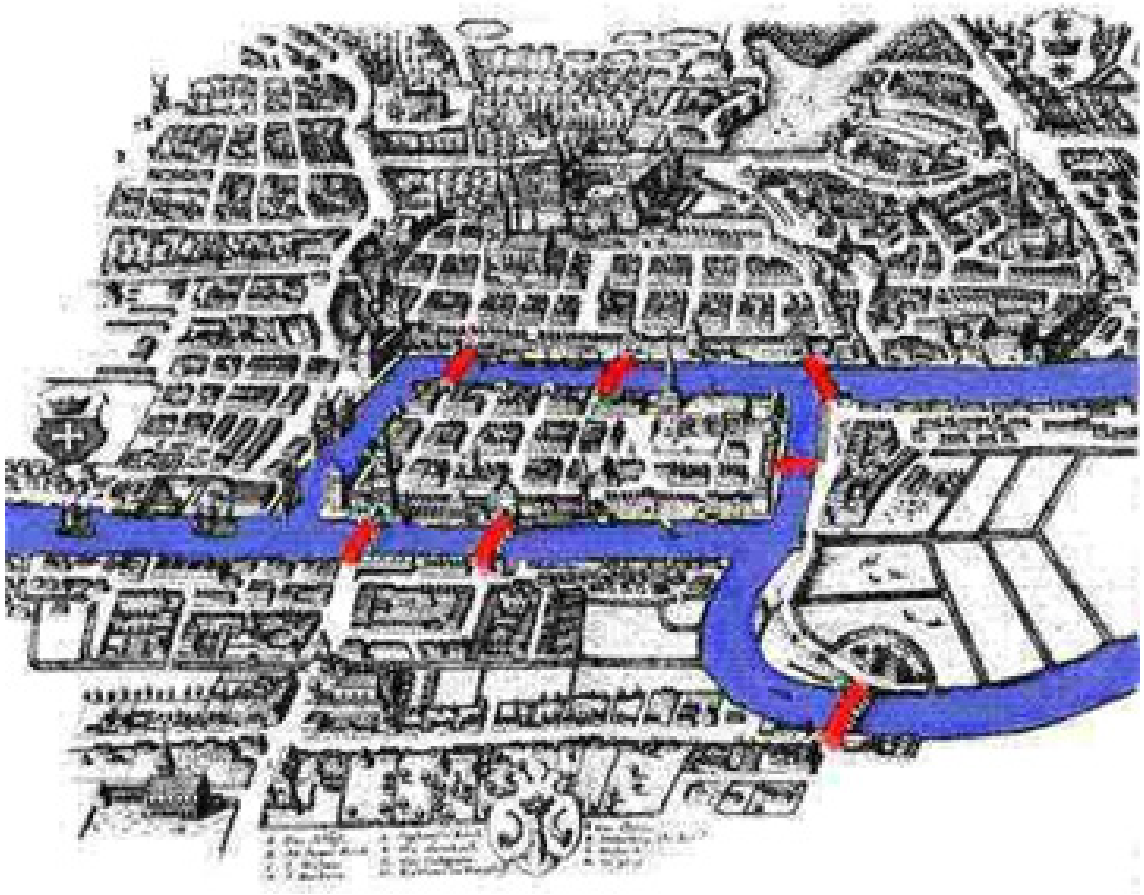

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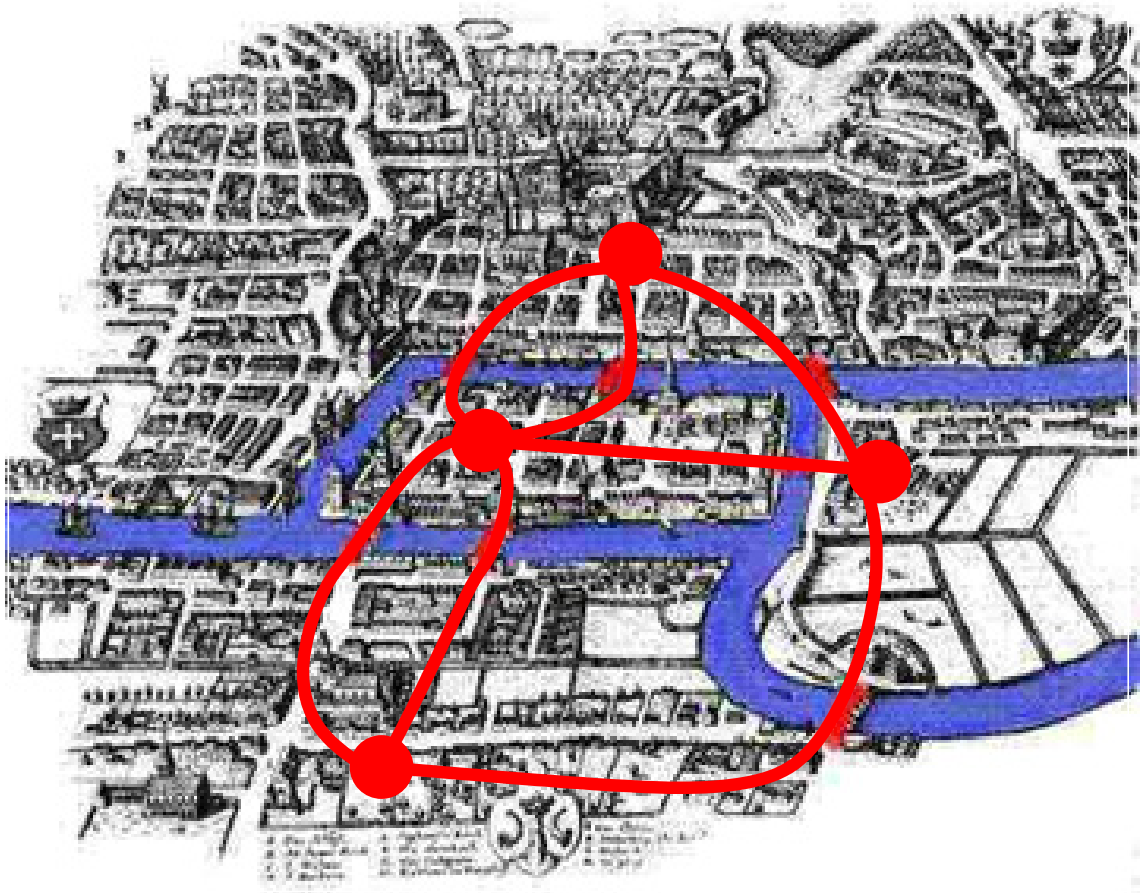

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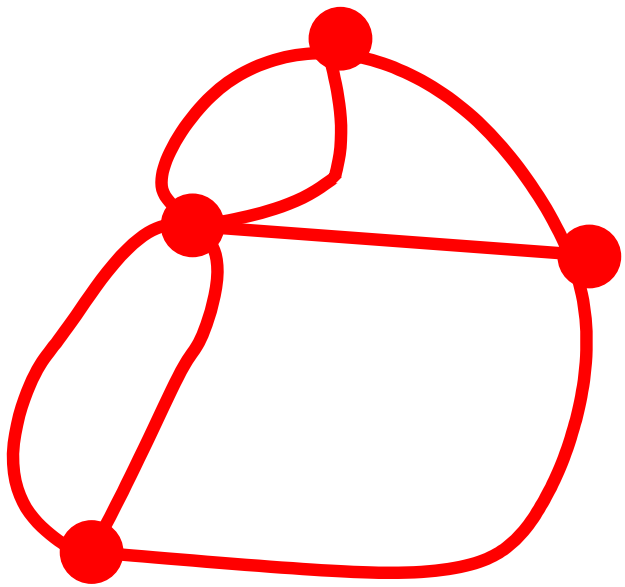
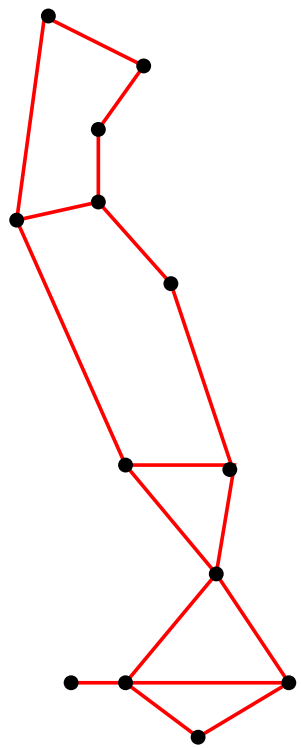


In 1736 Euler published a paper answering this question and founding graph theory

Representing Distances



- Consider some graph
- With weights representing the distance between nodes
- What is the shortest distance between S and I?

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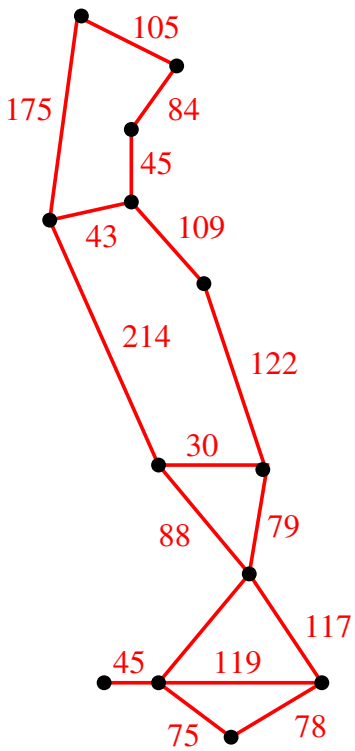

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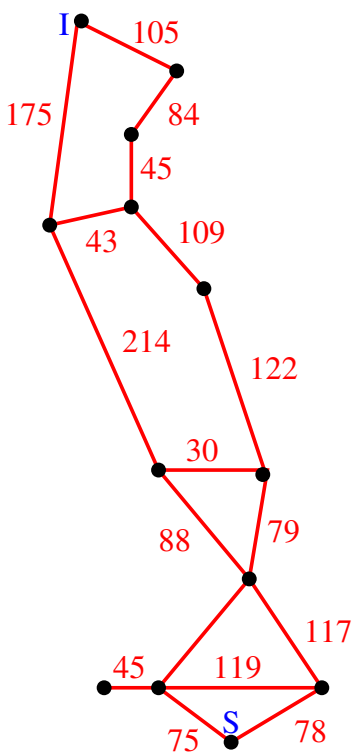

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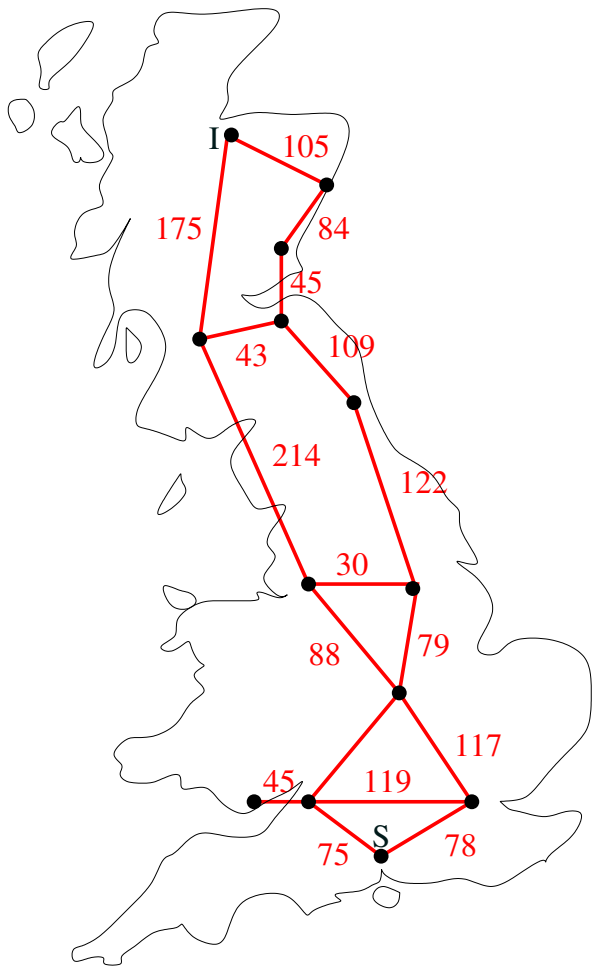

- Consider some graph
- With weights representing the distance between nodes
- What is the shortest distance between S and I?

Other Applications

- We could take the weights to represent the time taken to travel between nodes
- In a computer network the weights might represent the bandwidth
- In a representation of a transport system the weights might represent the carrying capacity of the traffic on a road
- Graphs can be used to represent other kinds of relationships
- E.g. We could create a digraph of links between web pages

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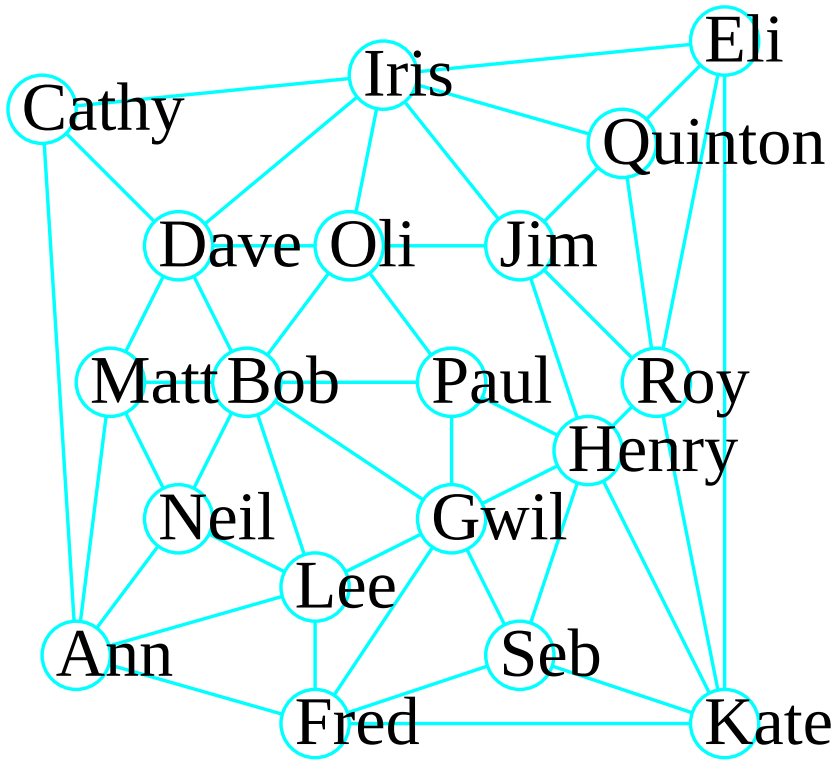
Christmas Card Problem

- I have four types of Christmas cards
- Some of my friends know each other



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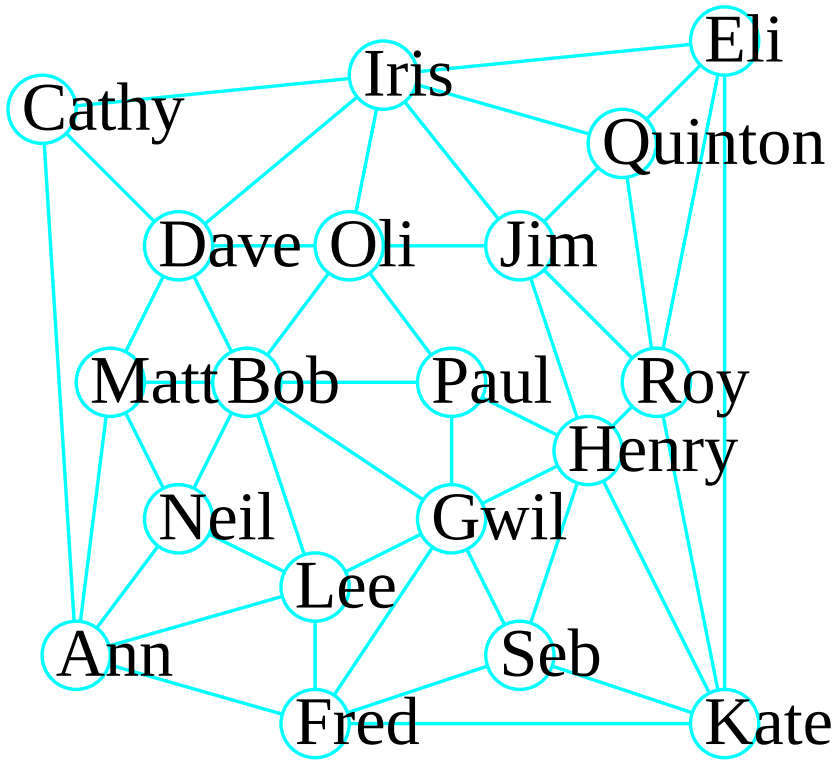

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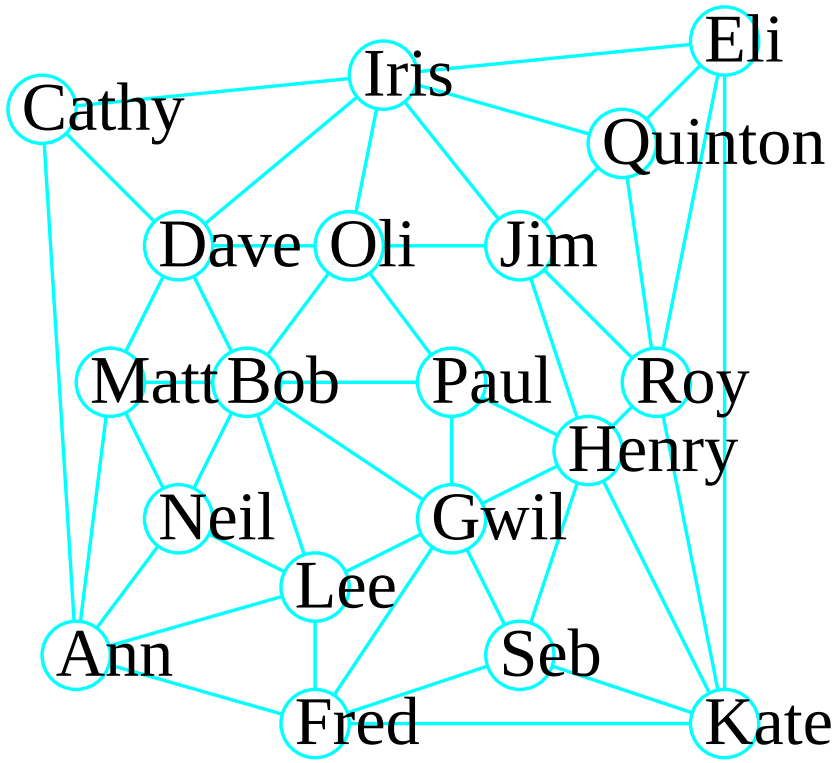

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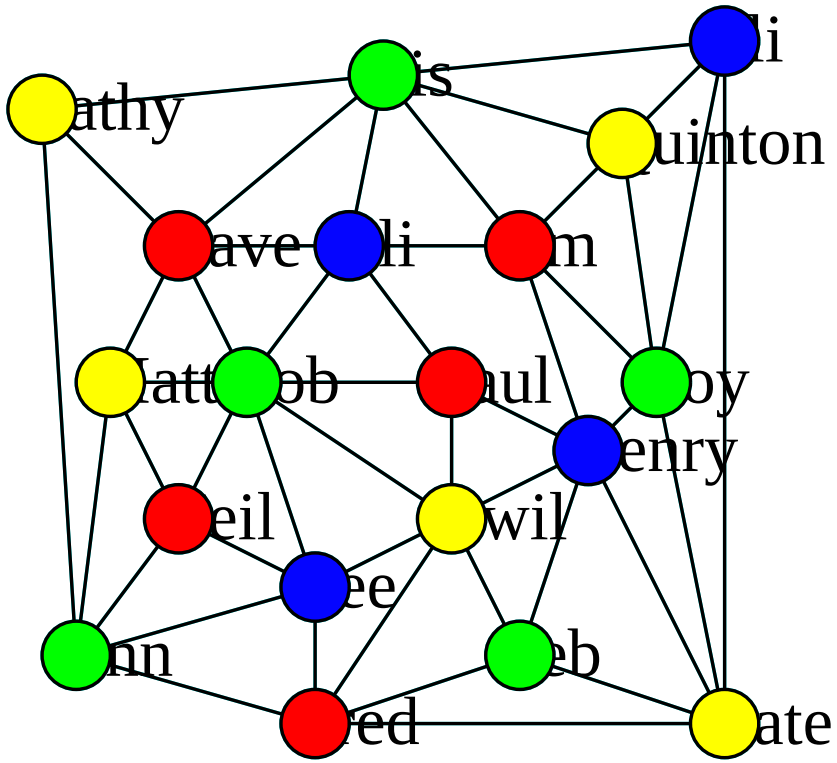

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A Real World Problem

- A food company used different colour bags for each of it products
- To save money they reduced the stock of bags to 25
- They wanted to know what items to put in what bags so that as few customers as possible would have items with the same colour bags
- This can again be reduced to a graph colouring problem
  - ⋆ Each node represents an item
  - ⋆ The edges were weighted by the number of customers that took both items
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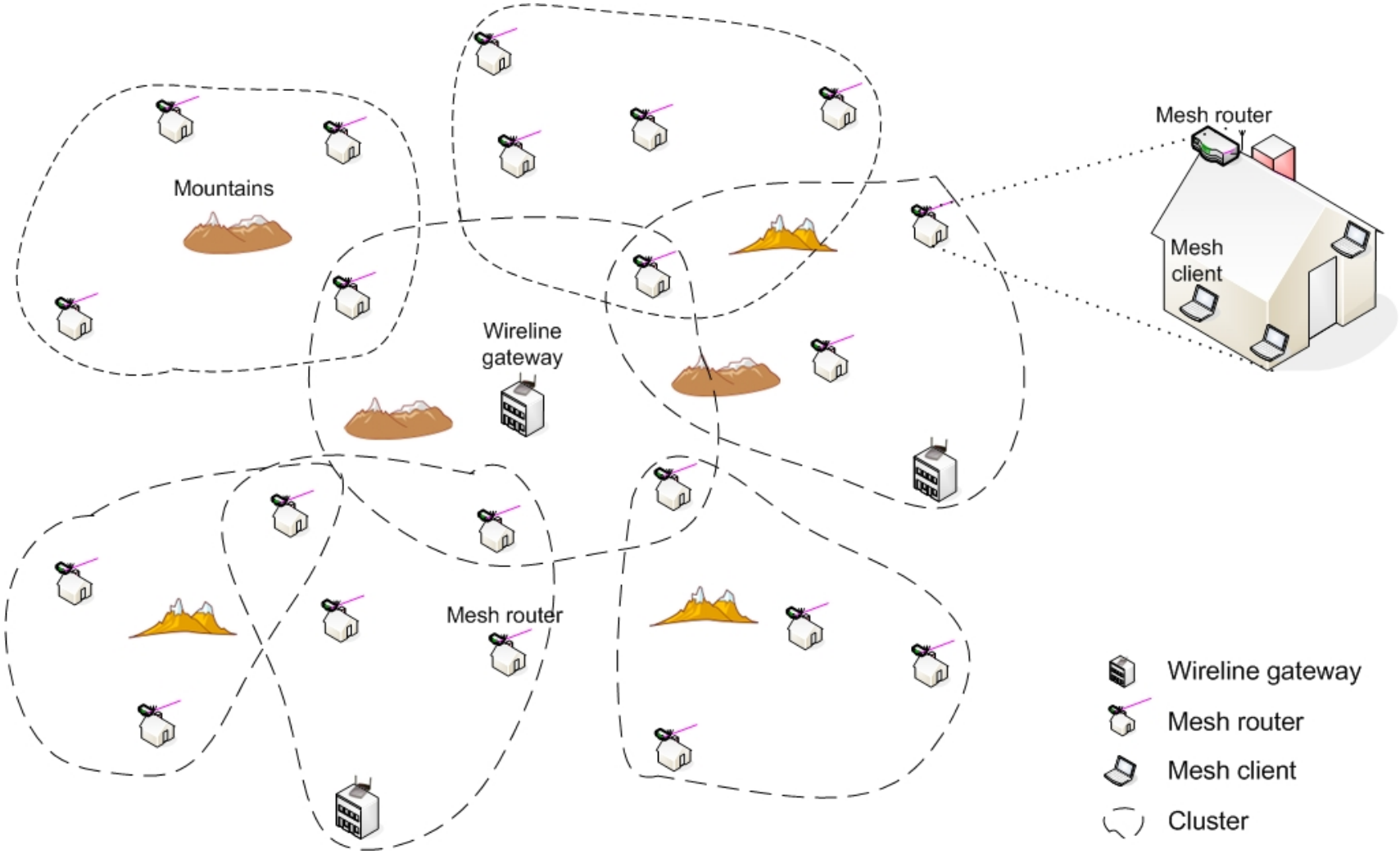
Frequency Assignment Problem



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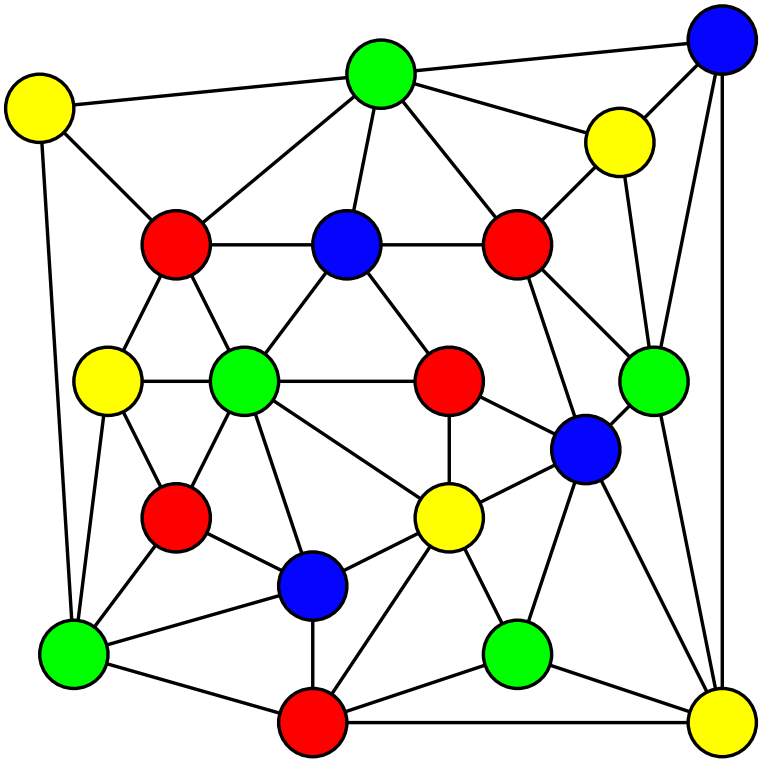

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  where $n = |\mathcal{V}|$

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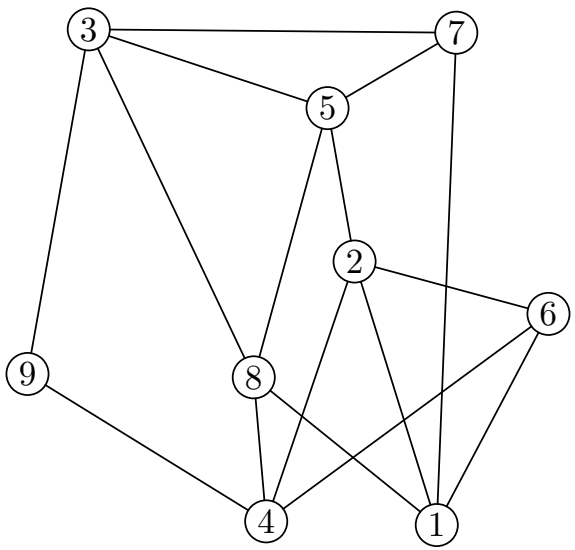
Representing Undirected Graphs



to

from

to	1	2	3	4	5	6	7	8	9
1	0	1	0	0	0	1	1	1	0
2	1	0	0	1	1	1	0	0	0
3	0	0	0	0	1	0	1	1	1
4	0	1	0	0	0	1	0	1	1
5	0	1	1	0	0	0	1	1	0
6	1	1	0	1	0	0	0	0	0
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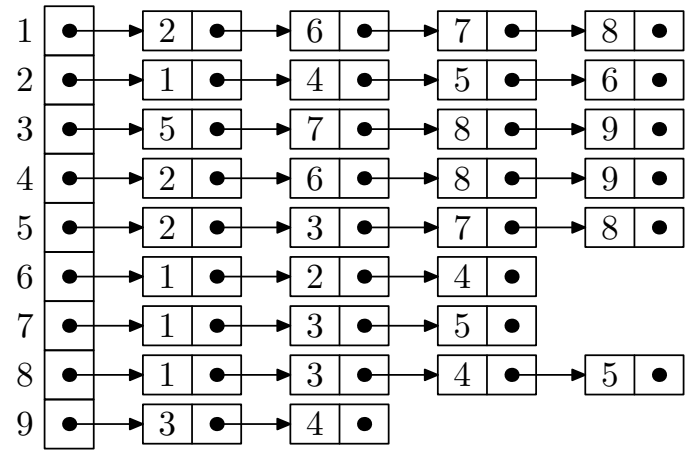
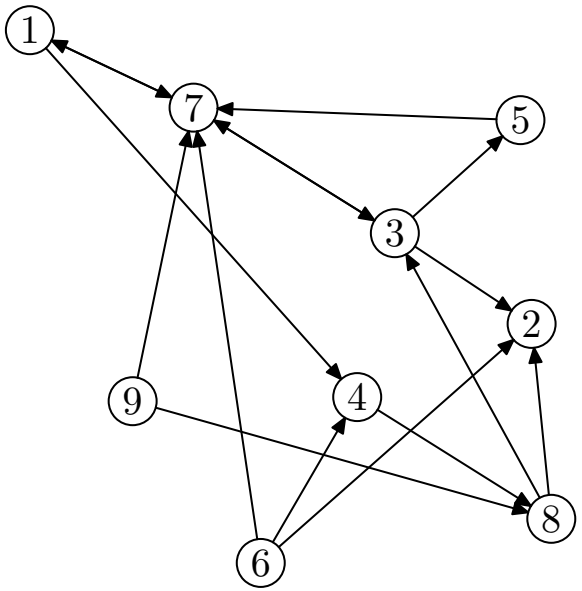


Graph

Adjacency Matrix

Adjancency List

Representing Digraphs



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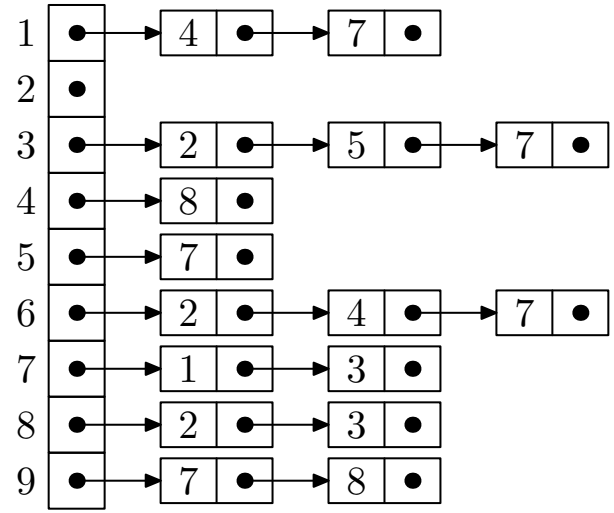

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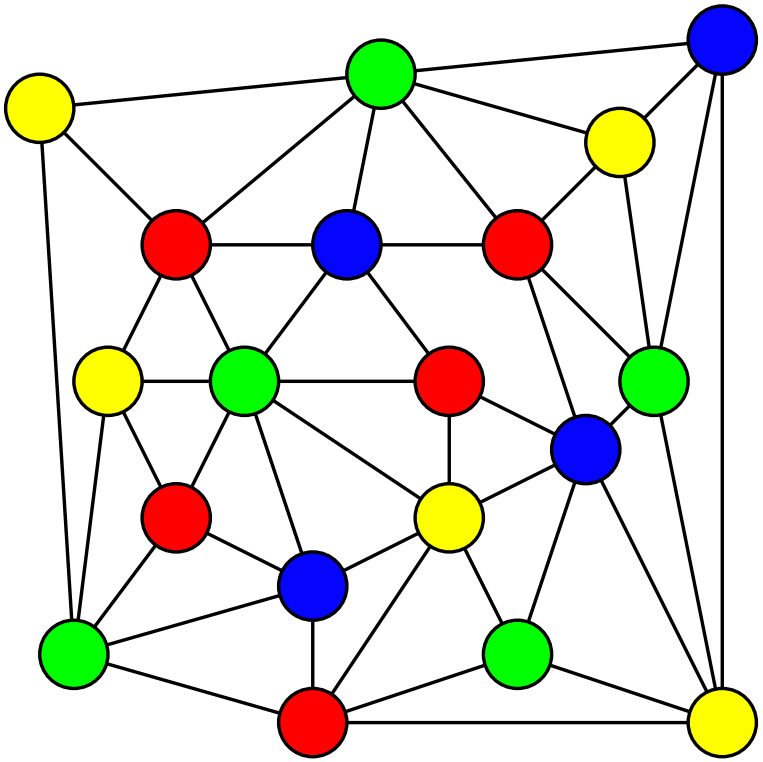
Outline

1. Graph Theory
2. Applications of Graphs
   - Geometric applications
   - Relational applications
3. Implementing Graphs
4. **Graph Problems**



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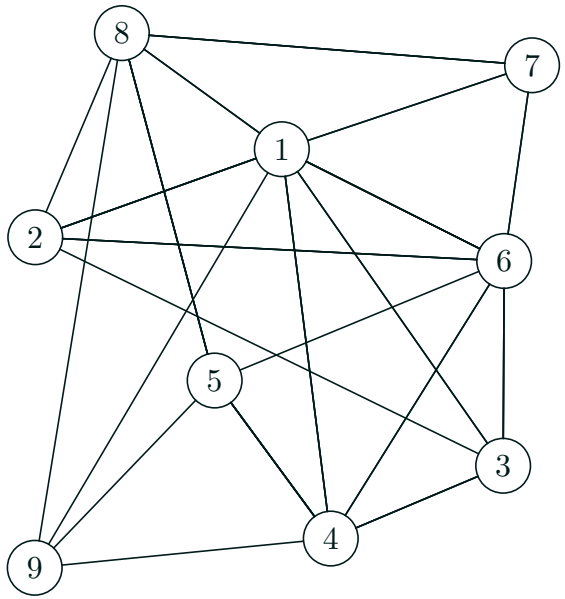
- The Hamilton cycle problem is to find a cycle that goes through each vertex exactly once



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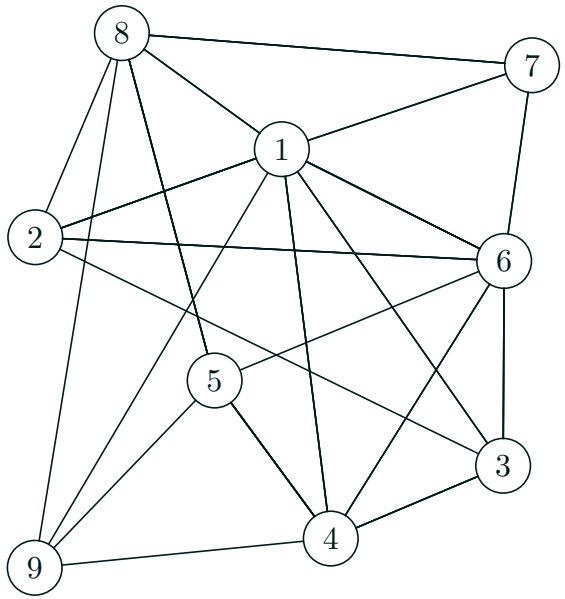

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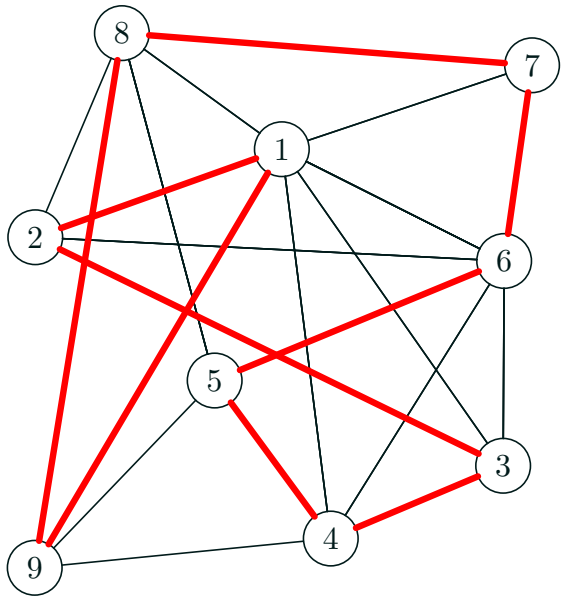

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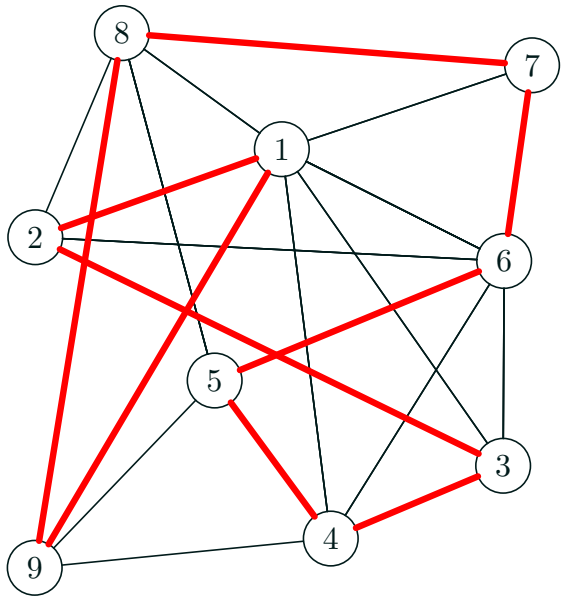

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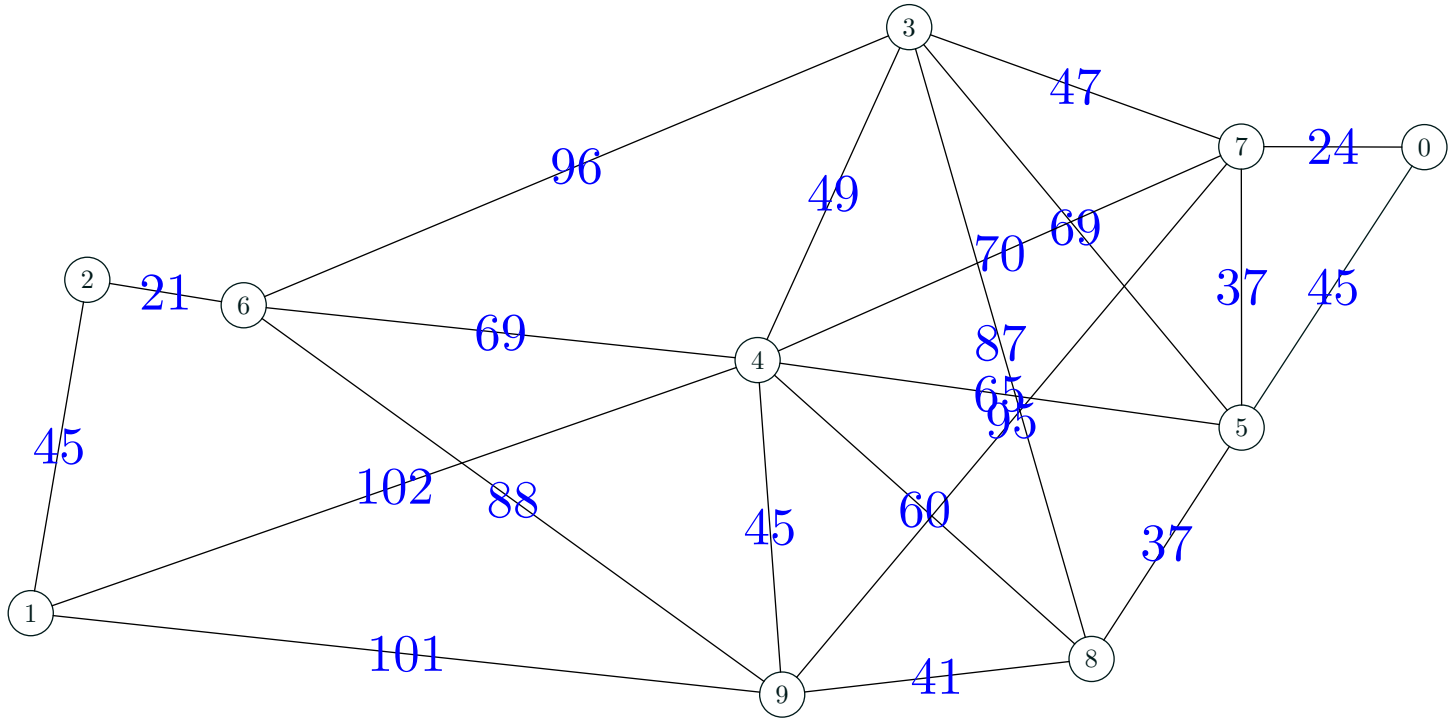

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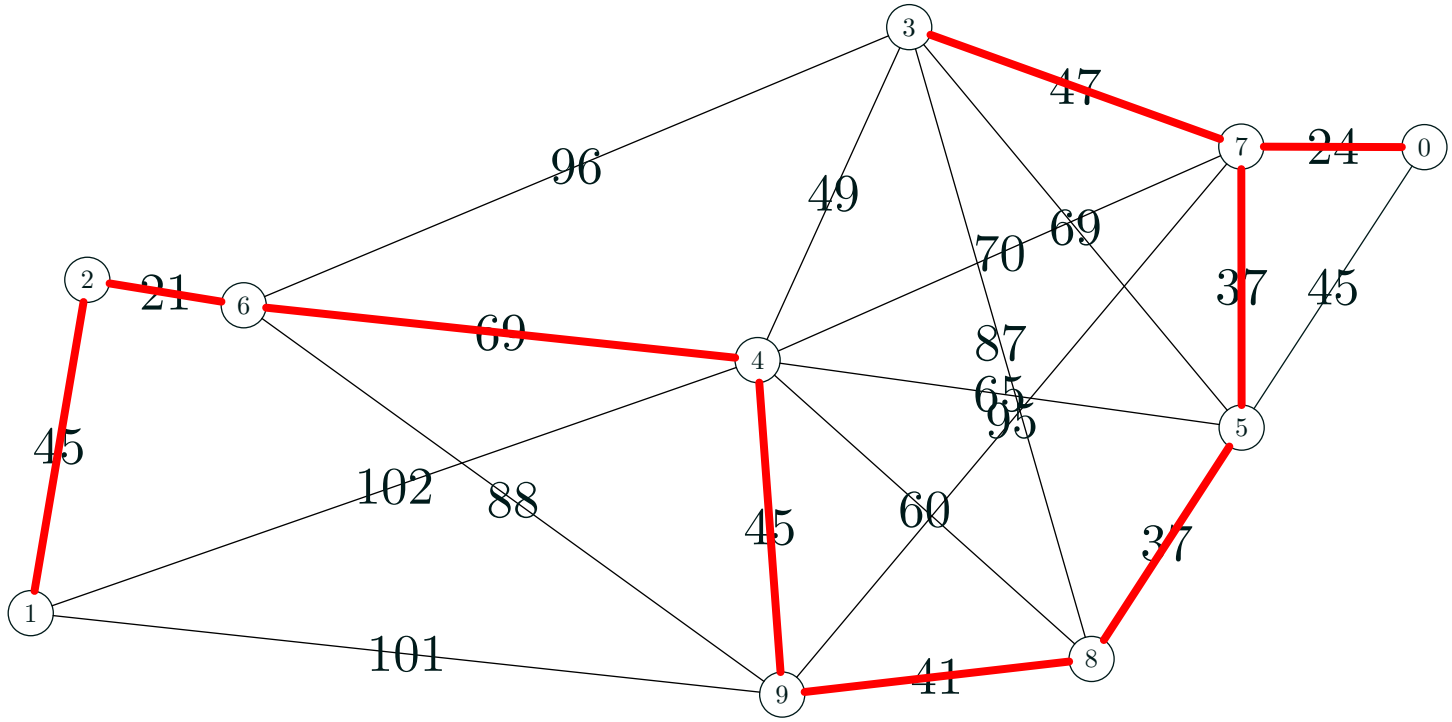

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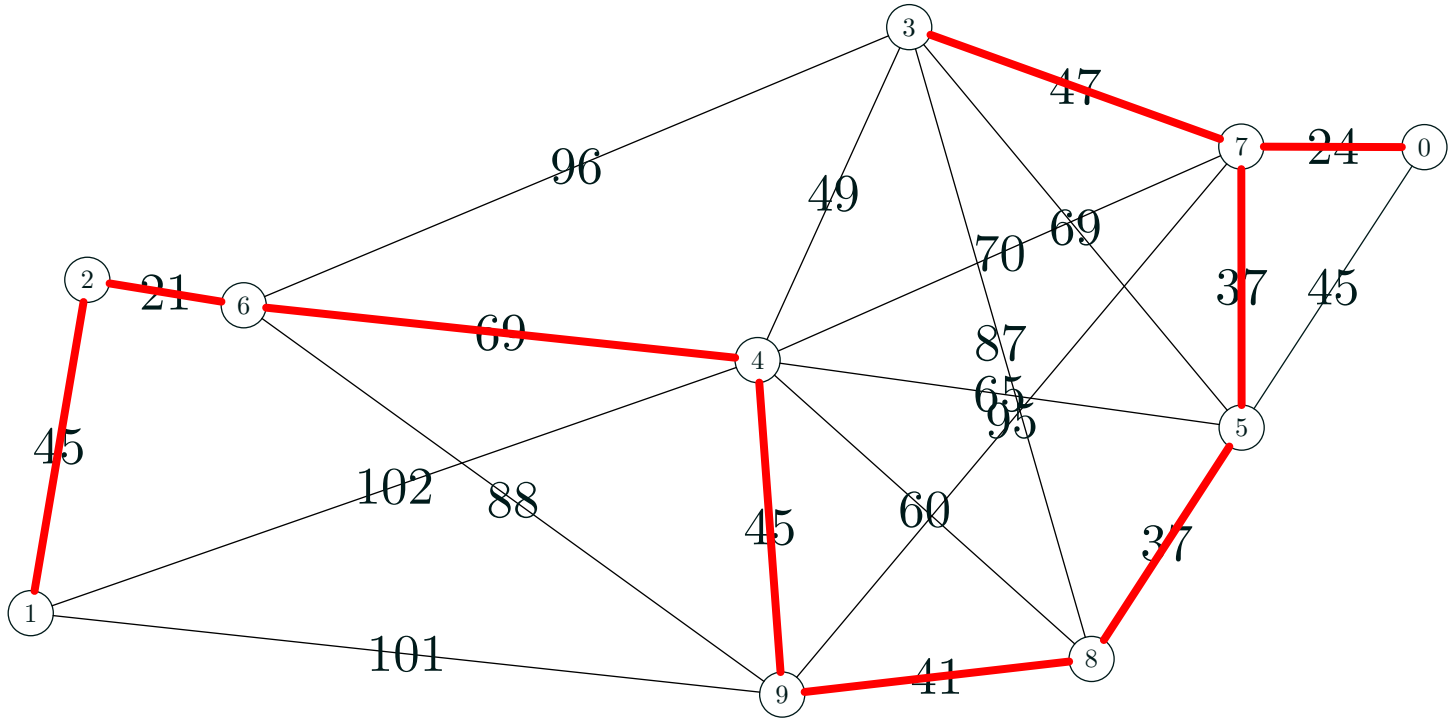

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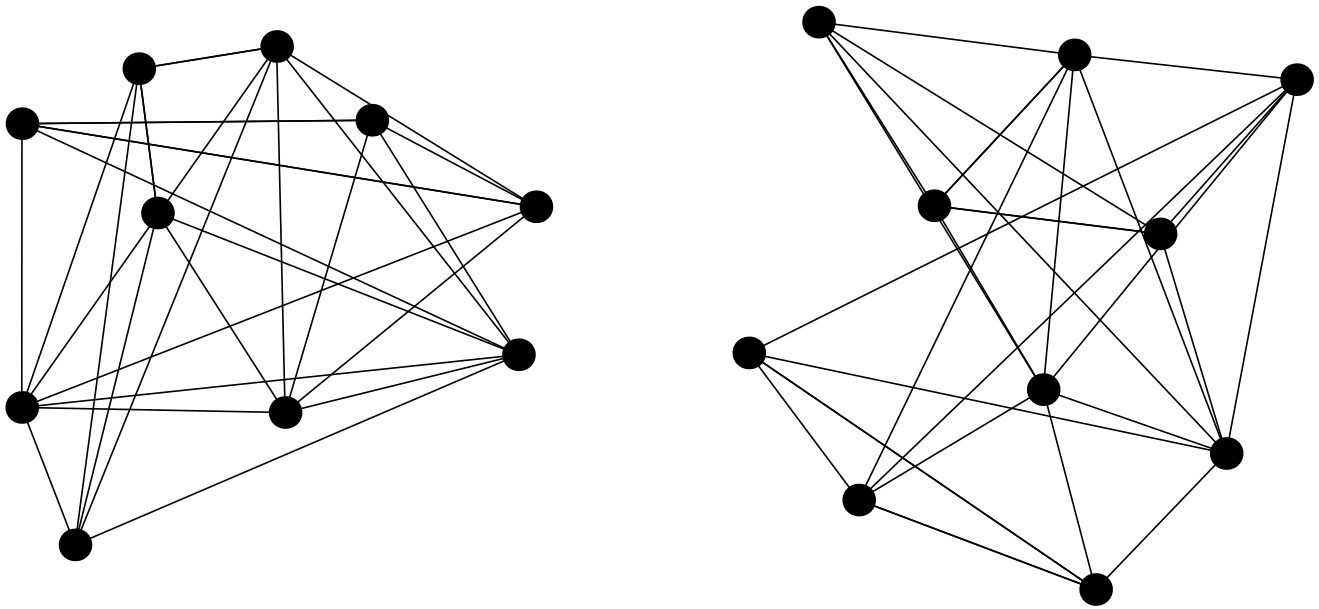

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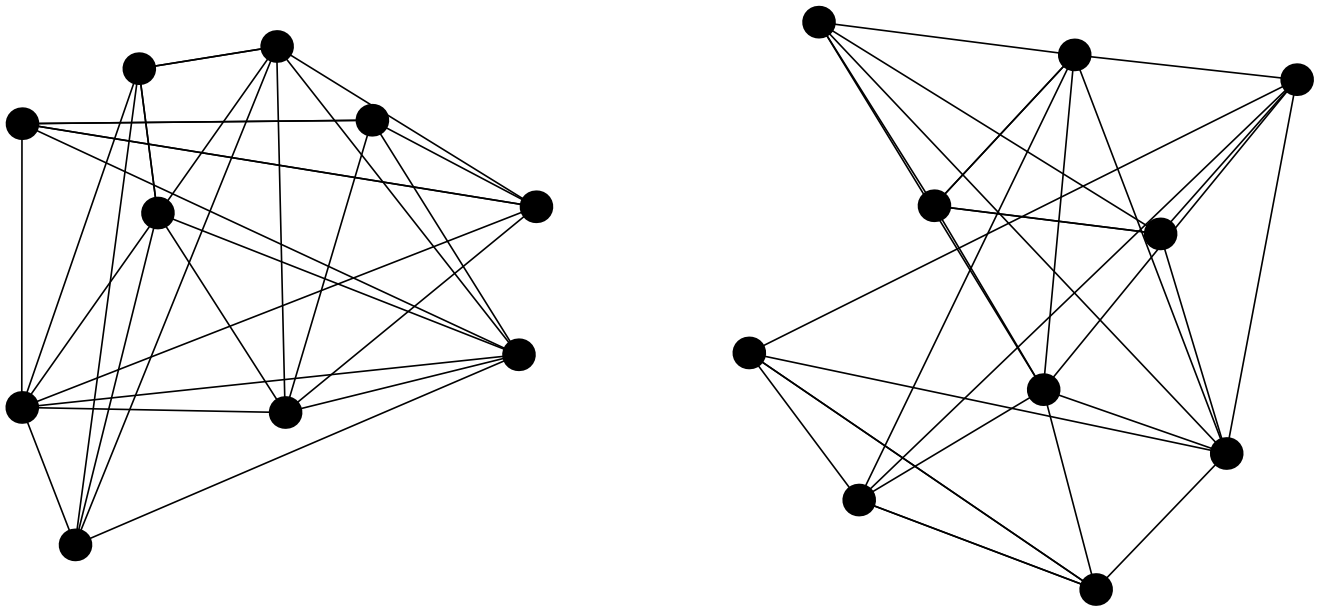

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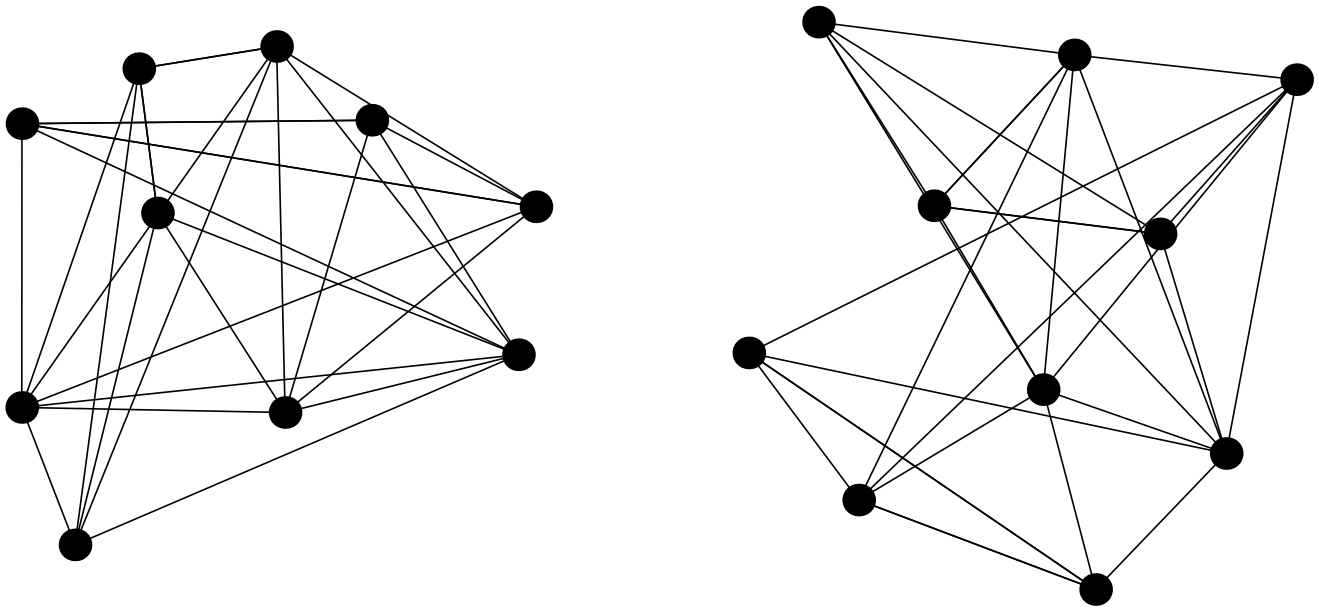

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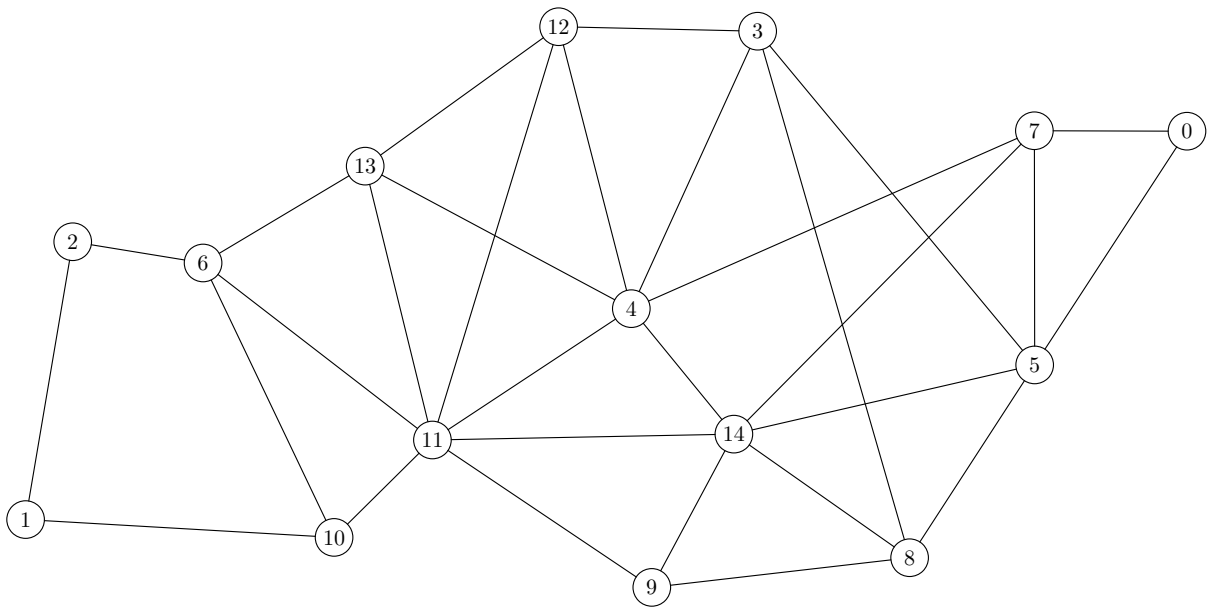

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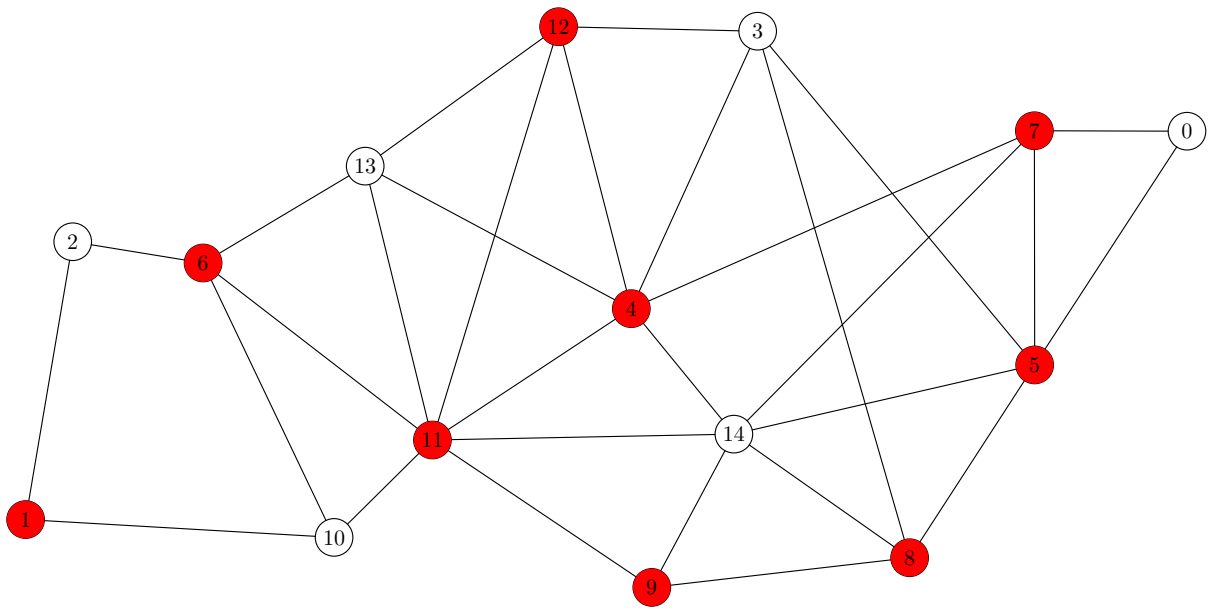

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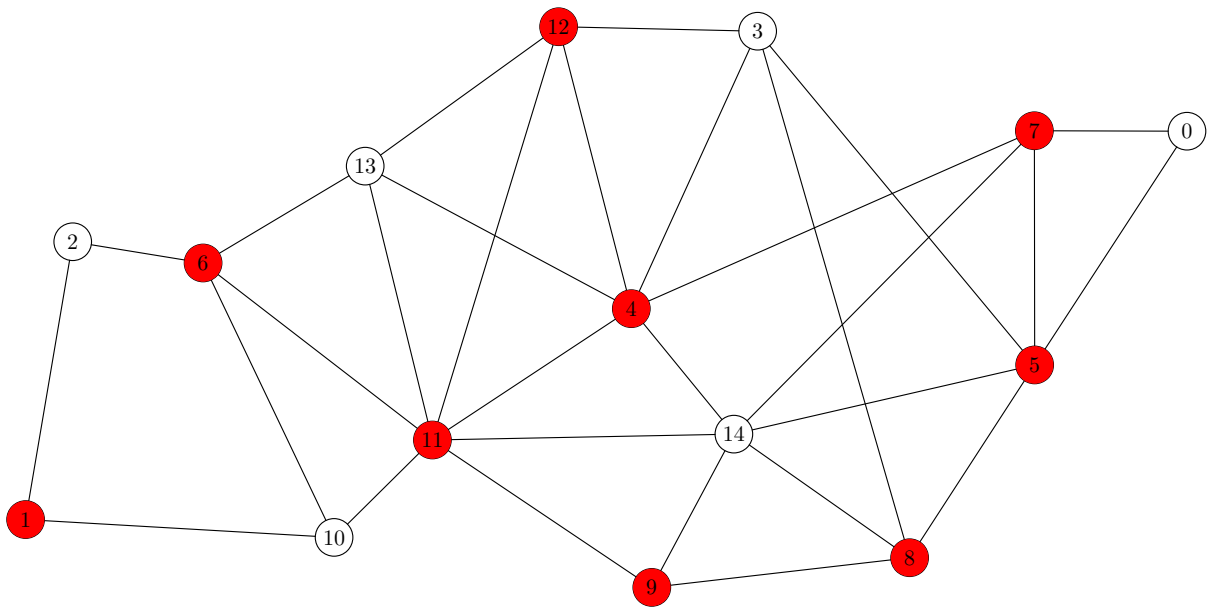

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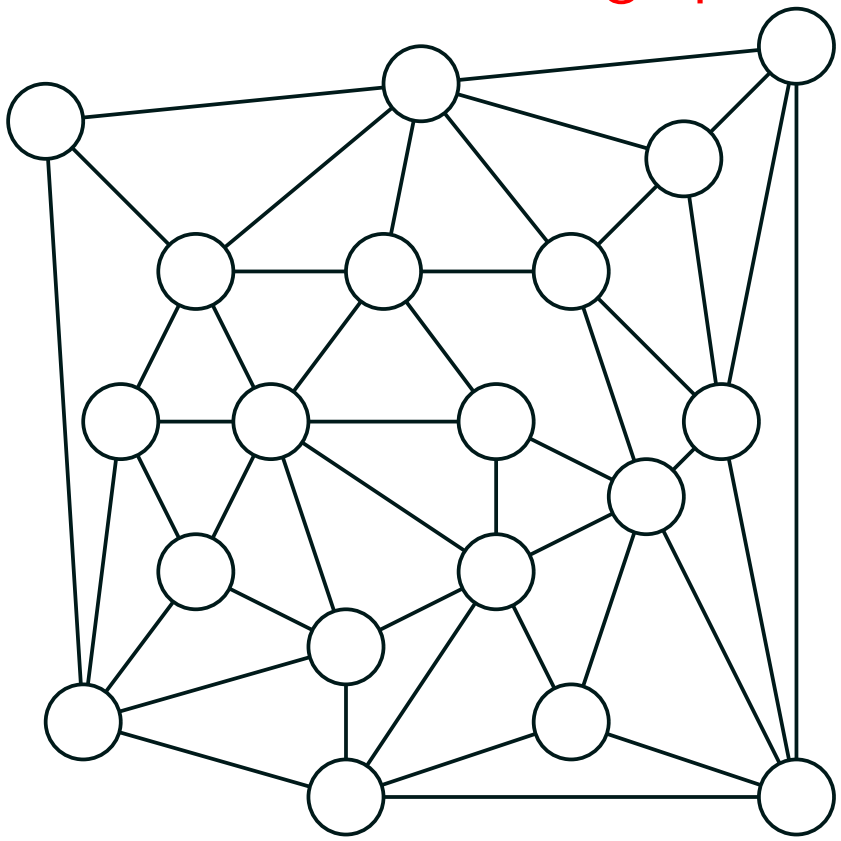

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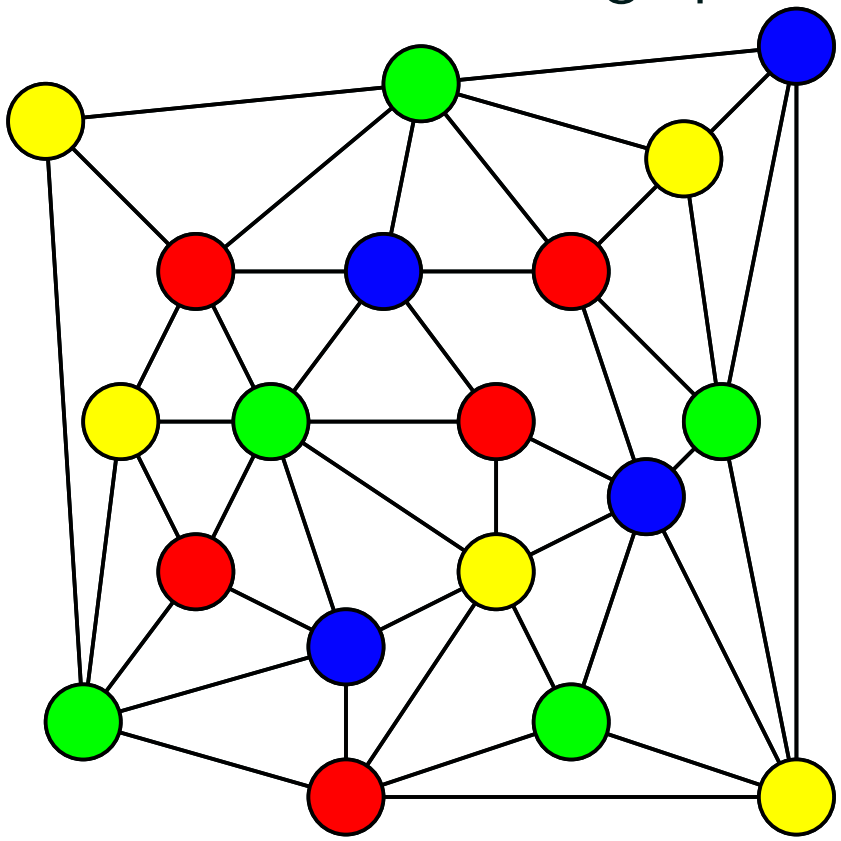

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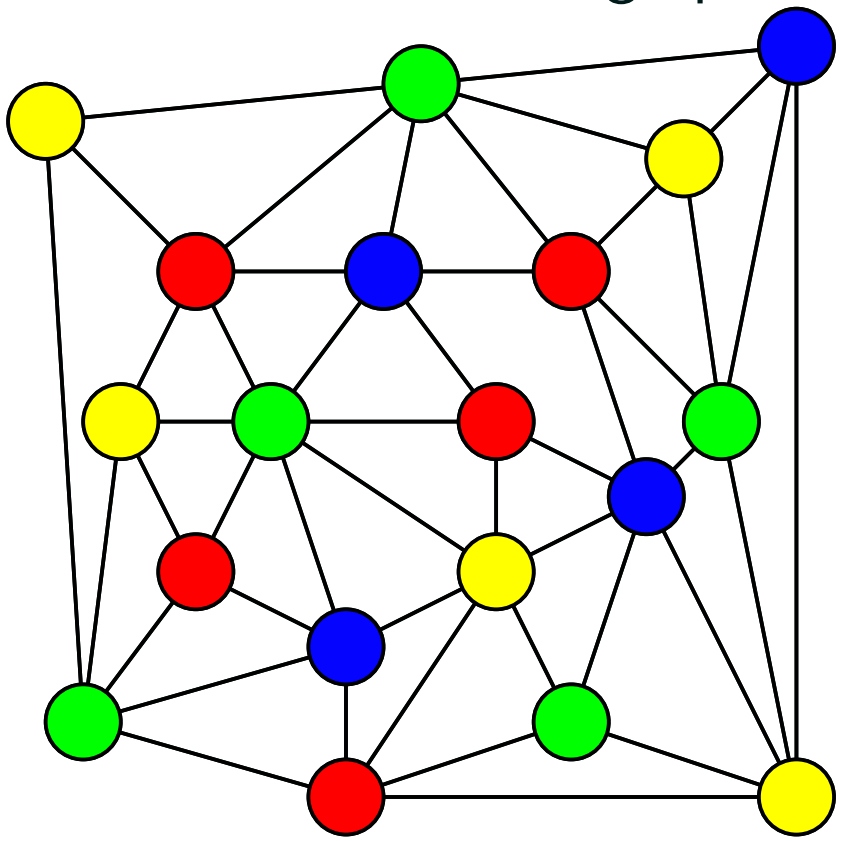

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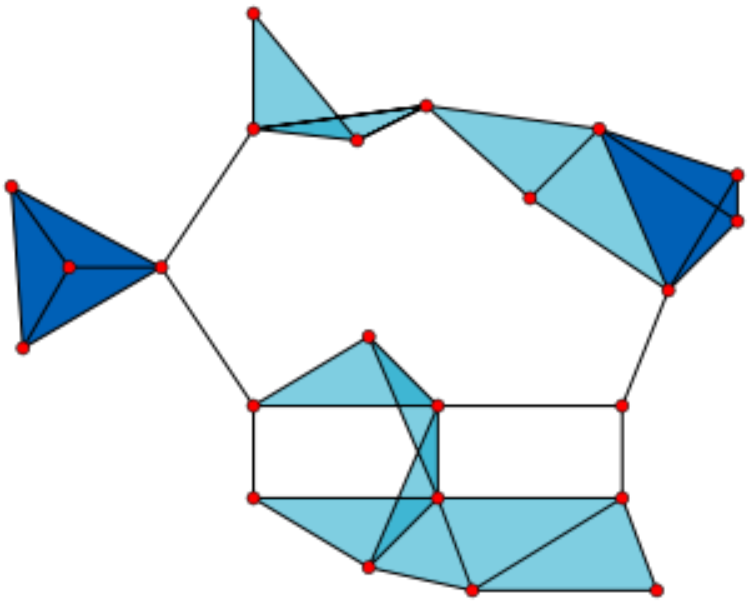

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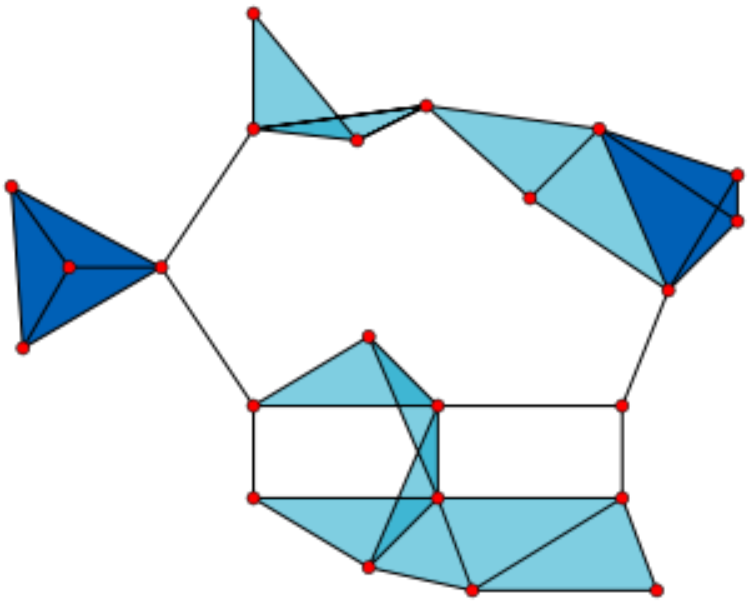

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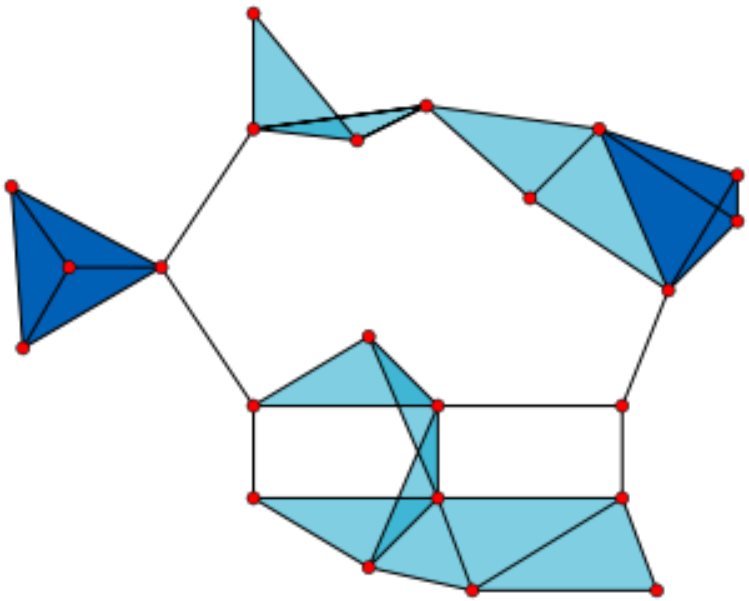

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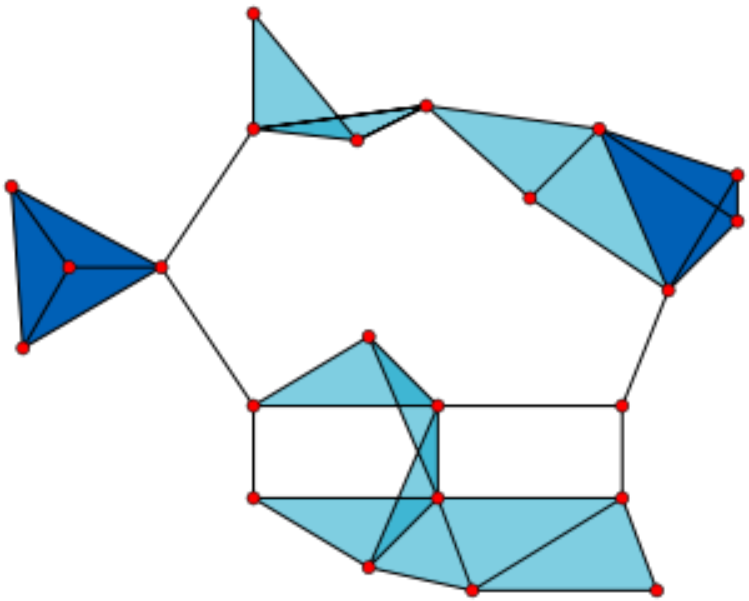

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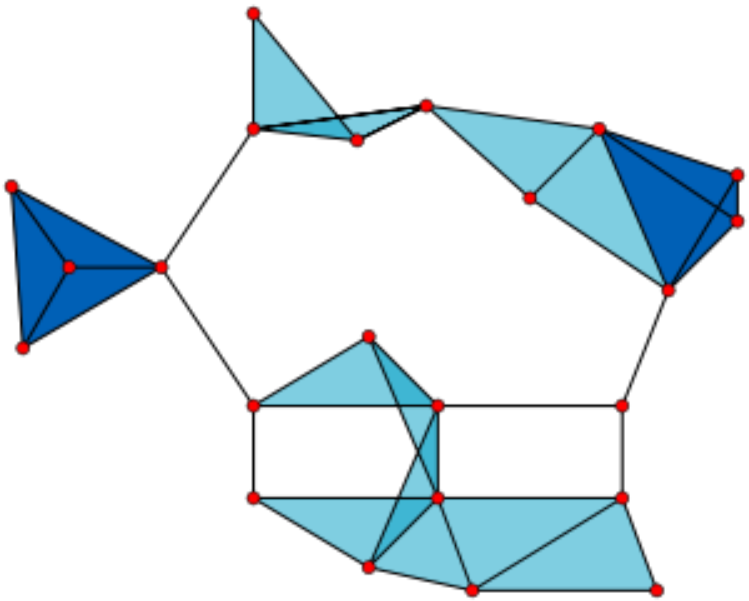

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- They appear in a huge number of disparate fields
- There are many problems for which efficient algorithms are known
- There are many problems which are believed to be hard—i.e. there aren't any efficient algorithms
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